

Presentation - Matgeo

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Problem Statement

The angle between the vectors $\hat{i} - \hat{j}$ and $\hat{j} - \hat{k}$ is

Solution

The cosine of the angle between two vectors $\mathbf{n}_1$ and $\mathbf{n}_2$

$$\cos\theta = \frac{\mathbf{n}_1^T \mathbf{n}_2}{\|\mathbf{n}_1\| \cdot \|\mathbf{n}_2\|} \quad (0.1)$$

Here,

$$\mathbf{n}_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \mathbf{n}_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \quad (0.2)$$

For the given two vectors, the value of their norms is $\sqrt{2}$ and

$$\mathbf{n}_1^T \mathbf{n}_2 = (1 \ -1 \ 0) \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \quad (0.3)$$

$$\mathbf{n}_1^T \mathbf{n}_2 = -1 \quad (0.4)$$

$$\cos\theta = \frac{-1}{(\sqrt{2})^2} = \frac{-1}{2} \quad (0.5)$$

Therefore the angle θ between the given vectors can be,

$$\theta = \frac{\pi}{3} \text{ or } \frac{2\pi}{3}$$

Plot

3D Vectors: $i - j$ and $j - k$

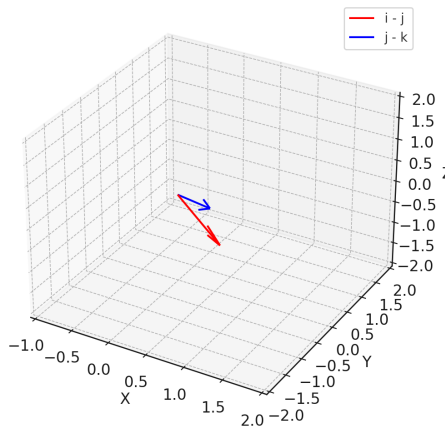


Figure: The Plot